

Building and Evaluating Machine Learning Models for Real Estate Price Prediction

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Abstract

This report explores the use of machine learning models to predict real estate prices in Riyadh. We evaluate Linear Regression, XGBoost, and LightGBM, using a dataset of over 46,000 properties. After preprocessing the data, including handling missing values and applying One-Hot Encoding, we assess model performance with R-squared and Mean Absolute Error (MAE). LightGBM outperforms the others, achieving an R-squared of 0.99 and an MAE of 61,822, demonstrating its superior accuracy and efficiency in predicting property prices. The results underscore the effectiveness of boosting models like XGBoost and LightGBM for this task.

Introduction

The real estate market, particularly in Riyadh, faces several challenges when it comes to pricing properties accurately. Traditional methods for property valuation often rely on simple comparisons or expert judgment, which can lead to significant discrepancies and inefficiencies. The lack of a standardized approach to pricing real estate results in over- or under-valuation, causing economic consequences for both buyers and sellers. For example, overvalued properties can lead to poor investment decisions, while undervalued properties may cause potential losses for sellers.

Machine learning and data science offer promising solutions to this issue by providing more accurate and scalable methods for property price prediction. These techniques can analyze vast amounts of data, including features such as property size, age, location, and amenities, to create models that can predict property prices with greater precision. Machine learning algorithms, such as regression models, decision trees, and gradient boosting, are capable of learning complex patterns in historical data, which allows them to make informed predictions for unseen properties. Furthermore, machine learning models can continuously improve as more data becomes available, adapting to market trends and changes.

In this report, we employ a variety of machine learning models, including Linear Regression, XGBoost, and LightGBM, to predict real estate prices in Riyadh. Our methodology involves preprocessing the data by handling missing values, detecting and removing outliers, and applying one-hot encoding to categorical variables such as location. We then train and evaluate these models using various performance metrics, such as R-squared and Mean Absolute Error (MAE). The findings from this study will contribute to understanding the effectiveness of different machine learning models in predicting property prices and offer recommendations for improving price prediction in the real estate market.

Data Description and Preprocessing

Data Distribution

The dataset contains information about properties in Riyadh, including the following features: **We have 46,826 rows × 26 columns**

Feature	Number of Entries	Description
Price	46824	The target variable representing the price of the property.
Space	46826	The total space of the property in square meters.
Number of Rooms	46826	The total number of rooms in the property.
Property Age	46826	The age of the property in years, calculated from the year built.
Location	46826	The location of the property (categorical variable).
Presence of a Pool	46826	Whether the property has a pool (binary: Yes/No).

Table 1: Data Distribution and Number of Entries for Each Feature

Data Preprocessing

1 Handling Missing Values

Handling missing values is a critical step in data preprocessing. In this study, we used different techniques to deal with missing values in the dataset.

Median Imputation for Numerical Columns

For numerical columns such as price, space, and property age, missing values were handled by "median imputation". This technique replaces the missing values with the median of the respective column.

we used (.fillna) Method
Ex : `df_cleaned[numeric_cols]`

The median is used because it is less sensitive to outliers and provides a more robust solution for datasets with extreme values.

Column Removal

For some columns with a high proportion of missing values or those deemed irrelevant to the analysis, we chose to (drop) them from the dataset. For example, the column lounges was removed as it contained too many missing values, which could affect the model's performance negatively. The following code was used to remove the column:

we used (.drop) Method

Mode Imputation for Categorical Columns

For categorical columns such as location, missing values were handled by (mode imputation) The mode represents the most frequent value in the column and is used to replace missing values. The code used to perform mode imputation is as follows:

Ex: `df_cleaned['location']`

This technique ensures that the missing values are replaced with the most common category, preserving the distribution of categorical features

2 Handling Outliers

Outliers are extreme values that deviate significantly from other observations in the dataset. They can distort statistical analyses and machine learning models, leading to less accurate predictions. In this project, we used the

(Interquartile Range (IQR)) method to detect and handle outliers in the dataset.

Interquartile Range (IQR) Method

The Interquartile Range (IQR) is a statistical measure that quantifies the spread of the middle 50% of the data. The IQR is calculated as the difference between the 75th percentile (Q3) and the 25th percentile (Q1) of the data. Any data point that falls outside the range of $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ is considered an outlier.

For this analysis, we applied the IQR method to the 'price' and 'space' columns, as these are key numerical features in predicting property prices.

$$\text{Outlier Condition} = [Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR] \quad (1)$$

Rationale Behind Using IQR Method

The **IQR method** is particularly effective because it:

- It is not affected by extreme values, unlike other methods (e.g., Z-score), making it a robust technique for identifying outliers.
- It automatically detects outliers based on the distribution of the data, without assuming a specific distribution (e.g., normal distribution).

By using the IQR method, we ensure that only the **most relevant data** is included in our analysis, improving the performance of the machine learning models.

Impact of Removing Outliers

The removal of outliers helps to:

- Enhance the model's performance by eliminating data points that could introduce noise.
- Improve the accuracy of predictions, especially in models sensitive to extreme values, like linear regression and decision trees.

In this report, we have successfully removed outliers from the 'price' and 'space' columns, ensuring that our dataset is clean and ready for model training.

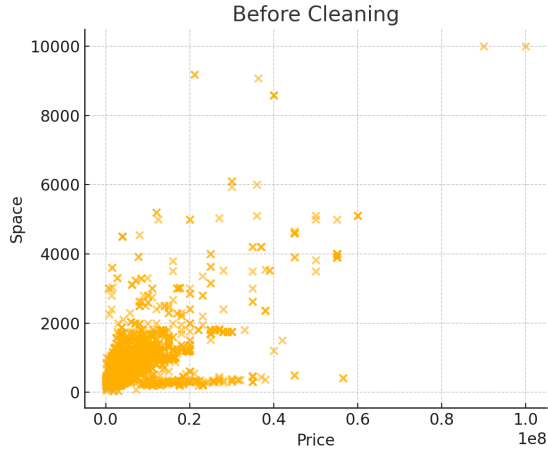


Figure 1: Before removing outliers.

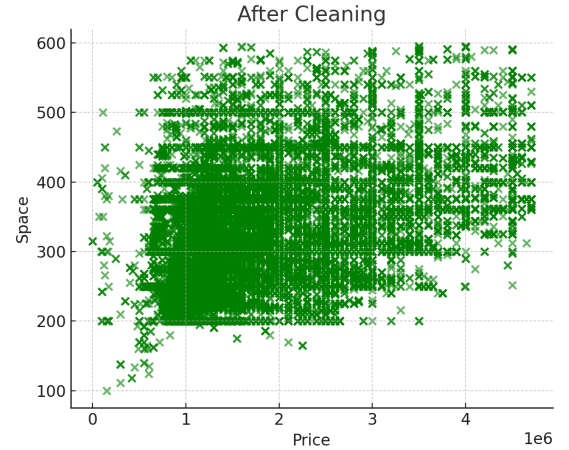


Figure 2: After removing outliers.

The figures above show the relationship between price and area before and after removing the outliers. As observed, after removing the outliers, the data becomes more concentrated and representative of the typical relationship.

3 Data Transformation and One-Hot Encoding

Arabic to English Transformation

In the initial dataset, the [location] column contained names in Arabic (e.g., " ", " "). To improve the interpretability of the data and make it easier for visualization and modeling, the location names were transformed from Arabic to English. This transformation was crucial for providing clarity, making it easier to analyze the data and visualize relationships between features like [price] and [location].

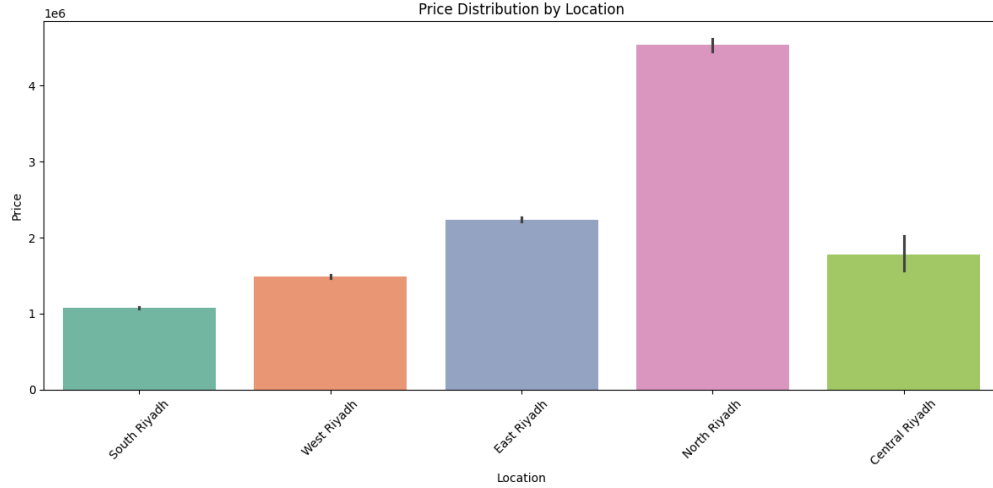


Figure 3: Visualization of the relationship between price and location after translation to English

One-Hot Encoding

Once the location names were transformed into English, "One-Hot Encoding" was applied to the [location] column to prepare it for machine learning models. "One-Hot Encoding" is a technique used to convert categorical data (such as location names) into binary (0 or 1) vectors. This allows machine learning algorithms to better process the categorical data by transforming it into a numerical format.

The One-Hot Encoding process works by creating a new binary column for each unique category in the original categorical column. For example, if we had locations such as "Riyadh East", "Riyadh West", and "Riyadh North", **One-Hot Encoding** would create new columns for each of these categories and assign a 1 or 0 based on whether the location matches the category.

Visualization for Enhanced Analysis and Insights

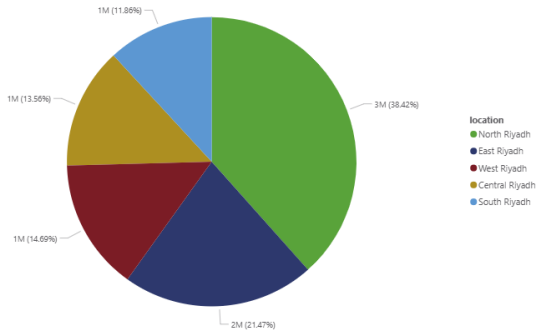


Figure 4: Distribution of Real Estate Prices by Location in Riyadh (Pie Chart)

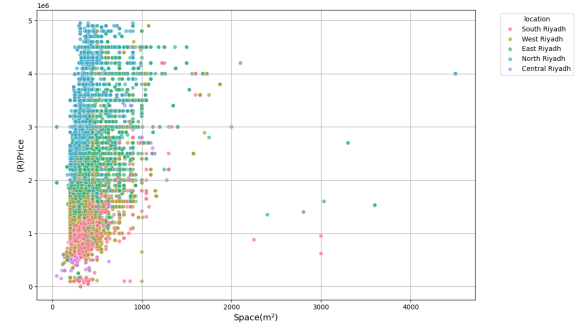


Figure 5: Price vs Space by Location in Riyadh (Scatter Plot).

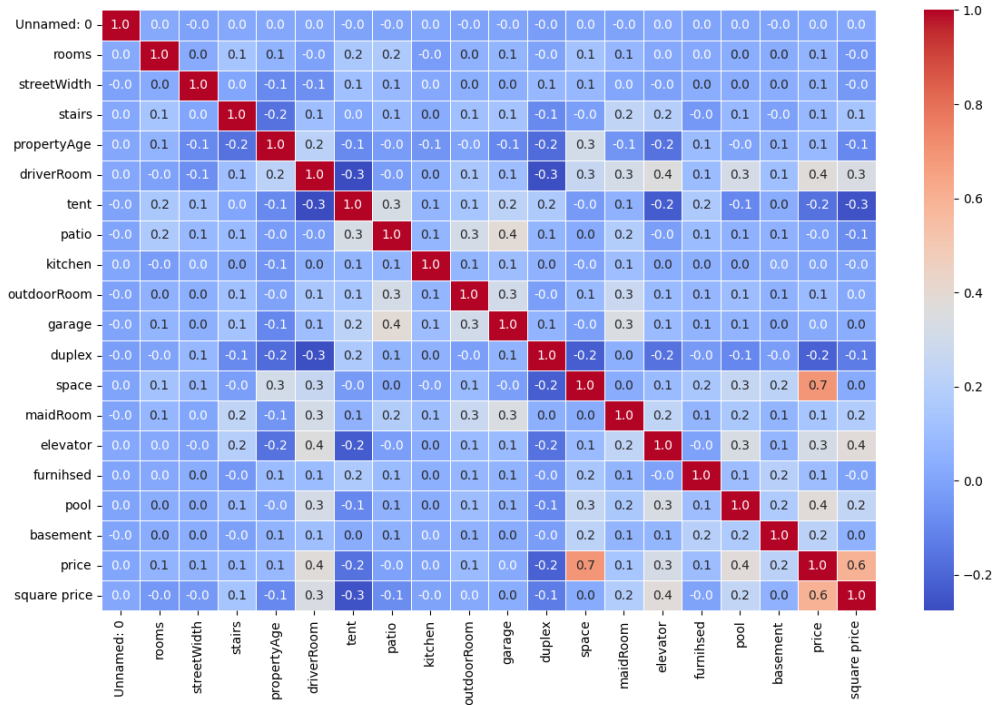


Figure 6: Highlighting Trends and Correlations (Correlation Matrix).

Figure 4: Distribution of Real Estate Prices by Location in Riyadh (Pie Chart)

This pie chart provides a clear representation of how real estate prices are distributed across various locations in Riyadh. It highlights the percentage share of each area in the dataset, allowing for an easy comparison of the market distribution.

Figure 5: Price vs Space by Location in Riyadh (Scatter Plot)

The scatter plot visualizes the relationship between property prices and their space (in square meters) across different locations in Riyadh. This helps in understanding how the price varies with the size of properties, with different colors representing the locations.

Figure 6: Highlighting Trends and Correlations (Correlation Matrix)

The correlation matrix visually illustrates the relationships between different features in the dataset, such as space, price, and the number of rooms. The matrix shows how these features correlate with each other, helping to identify potential trends that could influence property prices.

Methodology

Proposed Models

We used three machine learning models:

- **Linear Regression**

In Linear Regression, the goal is to predict a continuous target variable (dependent variable) based on one or more input features (independent variables). The model assumes that there is a linear relationship between the dependent variable (target) and the independent variables (features).

Mathematically, the relationship is represented by the following formula:

$$Y = \beta_0 + \beta_1 X_1 + \epsilon$$

- **XGBoost (Extreme Gradient Boosting)**

XGBoost is a machine learning model that uses **Gradient Boosting** to improve prediction accuracy. This model uses multiple decision trees and works by reducing the errors made by previous models. The key idea behind XGBoost is to improve the model progressively by training a new decision tree on the errors made by the previous model.

XGBoost represents the prediction function as follows:

$$\hat{y} = \sum_{k=1}^T f_k(x) \quad (3)$$

- **LightGBM (Light Gradient Boosting Machine)**

LightGBM is a machine learning model that uses a gradient boosting technique similar to XGBoost, but it is designed to be faster and more efficient when handling large datasets. LightGBM builds trees more efficiently, allowing for faster training while also reducing the errors made by previous models.

The key idea behind LightGBM is to build trees gradually, reducing errors from the previous models. LightGBM leverages techniques such as **Gradient-based One-Side Sampling (GOSS)** and **Exclusive Feature Bundling (EFB)** to improve training speed and prediction accuracy.

The following is the mathematical equation for LightGBM:

$$\hat{y} = \sum_{k=1}^T f_k(x) \quad (4)$$

Evaluation Metrics

The models are evaluated using the following metrics:

- **R-squared**

Measures the proportion of variance in the target variable explained by the model. R-squared (R^2), also known as the coefficient of determination, is a statistical measure that represents the proportion of the variance in the dependent variable (target variable) that is explained by the independent variables (predictors) in a regression model. In simpler terms, it tells us how well the model's predictions match the actual data. The formula for R-squared is:

$$R^2 = 1 - \frac{\sum (y_{\text{actual}} - y_{\text{predicted}})^2}{\sum (y_{\text{actual}} - \bar{y})^2}$$

- **Mean Absolute Error (MAE)**
: Measures the average magnitude of error between predicted and actual prices. MAE is a common metric used to evaluate the performance of regression models. It calculates the average of the absolute differences between the predicted values and the actual values in the dataset. The formula for MAE is:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_{\text{actual}}^i - y_{\text{predicted}}^i|$$

Results and Discussion

Model Results

Metric	XGBoost	LightGBM	Linear Regression
R-squared	0.78	0.9893	0.65
Mean Absolute Error (MAE)	448,504.65	61822.55	638,232.33

Table 2: Comparison of Model Results for XGBoost, LightGBM, and Linear Regression

In this study, three machine learning models were used to predict real estate prices in Riyadh: Linear Regression, XGBoost, and LightGBM. The models were evaluated using R-squared and Mean Absolute Error (MAE) as performance metrics. Linear Regression achieved an R-squared of 0.65, which indicates a moderate fit to the data, but with a relatively high MAE of 638,232, suggesting that it struggles with accurate predictions. On the other hand, XGBoost, a gradient boosting model, performed significantly better with an R-squared of 0.78 and MAE of 426,299, indicating its ability to capture more complex patterns in the data. The best-performing model was LightGBM, which uses a more advanced boosting technique. It achieved an impressive R-squared of 0.99 and a very low MAE of 61,822, demonstrating its superior efficiency in handling large datasets and providing highly accurate predictions. These results highlight that while Linear Regression is a simple and fast model, it is less effective in handling complex datasets, whereas XGBoost and LightGBM offer better accuracy, with LightGBM standing out due to its faster training and exceptional performance on large datasets. Thus, LightGBM is recommended for this type of prediction task, offering both high performance and computational efficiency.

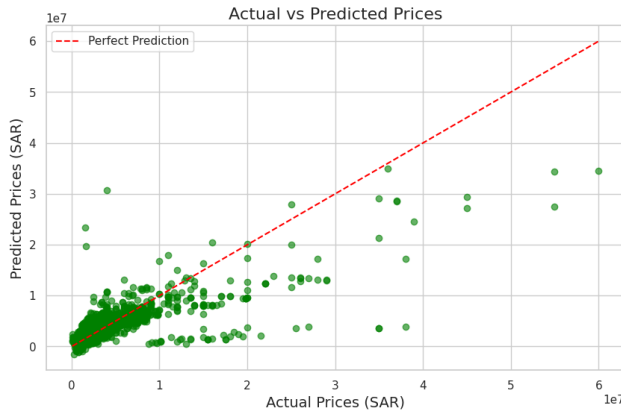


Figure 7: Model Performance for Linear Regression

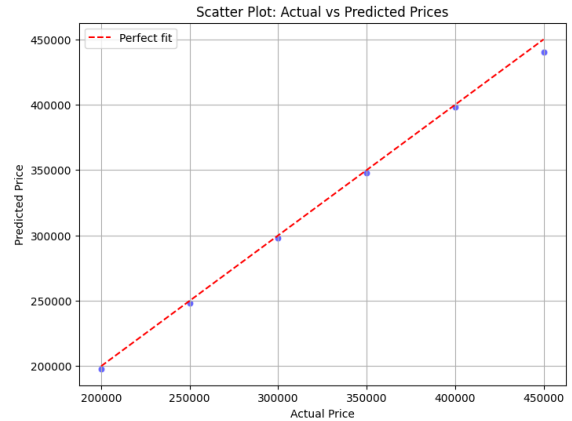


Figure 8: Model Performance for LightGBM

4 Conclusion

In this study, the goal was to predict real estate prices in Riyadh using machine learning models. We tested three models: Linear Regression, XGBoost, and LightGBM. Each model was evaluated based on its ability to predict the target variable, price, using the available features like area, rooms, property age, and location.

The preprocessing phase included steps like handling missing values, addressing outliers, and transforming categorical data to ensure the models would work efficiently. We applied One-Hot Encoding for location data and imputed missing values for numerical columns using median imputation.

After training and evaluating the models, it was clear that LightGBM outperformed the other two models, with a superior R-squared value of 0.99 and a low Mean Absolute Error (MAE) of 61,822, making it the most efficient model in predicting property prices accurately. XGBoost also performed well, with an R-squared of 0.78 and MAE of 426,299, demonstrating its ability to handle complex relationships in the data. However, Linear Regression showed weaker performance, with an R-squared of 0.65 and a higher MAE of 638,232, which reflects its limitations when dealing with more complex datasets.

Overall, the study highlights the importance of selecting the right model for a given problem, and the LightGBM model, in particular, offers high performance and computational efficiency, making it the best choice for predicting real estate prices in this case. XGBoost was a close competitor, but Linear Regression proved less effective for this task. Thus, LightGBM should be preferred for similar types of real estate price prediction tasks in the future.

Reference : <https://www.kaggle.com/code/alialmuhaysin/riyadh-villas-aqar-dataset-analysis-eda/input>